

Subject Description Form

Subject Code	EEE504
Subject Title	Wafer Fabrication and Laboratory
Credit Value	3
Level	5
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	The aim of this subject is to teach the basic concepts of wafer fabrication and measurement methodology and subsequently apply these concepts in laboratory exercises. Students will have a deeper understanding of the fabrication processes through hands-on training. Finally, the student will apply the knowledge and capability gained through these lab exercises to their laboratory operation and experimental setup demonstration.
Intended Learning Outcomes <i>(Note 1)</i>	Upon completion of the subject, students will be able to: Category A: Professional/academic knowledge and skills a. Understand and describe the stages of the wafer fabrication process, including the essential processes, wafer testing and characterization. b. Understand different types of material properties and coatings, and also study different measurement and characterization methods using different equipment (AFM, SME, etc.). c. Understand the characterization of microelectronic components (i.e., circuits and waveguides, filter and modulators); learn to control different instruments and setups using software, etc. Category B: Attributes for all-roundedness d. Communication effectively. e. Writing of the laboratory logbook and reports. f. Think critically and creatively.
Subject Synopsis/ Indicative Syllabus <i>(Note 2)</i>	Wafer fabrication process: understanding of fundamental wafer fabrication process: wafer preparation, spin coating, lithography process and electronic beam lithography. Different fabrication modules will be trained with process conditions and specifications, etc. Characterization and measurement: basic microelectronic measurement and analysis of transistor circuits, multi-stage amplifiers, operational amplifiers, current drivers and other semiconductor circuits, etc. Device evaluations: characteristic measurement of diodes, bipolar transistors, MOSFETs, waveguides and modulators. Build a testing setup and control different instruments with LabView software. Experiment metrology: consult scientific sources, analyze and display measurement results in tables and graphs, orally present scientific measurement results and write research reports and papers.

Teaching/Learning Methodology (Note 3)	The basic concepts of wafer fabrication and experiment characterization will be introduced in the lab. Students will understand the fabrication processes and the measurement of components/devices, and investigate the non-ideal characteristics of the components and device through testing. Students will be required to write a lab logbook and final report, which will deepen their understanding of microelectronic device design and fabrication. <table><tr><th rowspan="2">Teaching/Learning Methodology</th><th colspan="6">Intended subject learning outcomes</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th><th>e</th><th>f</th></tr><tr><td>Lectures</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td></td></tr><tr><td>Assignments & Exercises</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>Laboratory logbook</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr></table>	Teaching/Learning Methodology	Intended subject learning outcomes						a	b	c	d	e	f	Lectures	✓	✓	✓	✓	✓		Assignments & Exercises	✓	✓	✓	✓	✓	✓	Laboratory logbook	✓	✓	✓	✓	✓	✓												
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Assessment Methods in Alignment with Intended Learning Outcomes (Note 4)	<table><tr><th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">% weighting</th><th colspan="6">Intended subject learning outcomes to be assessed</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th><th>e</th><th>f</th></tr><tr><td>1. Assignment</td><td>50%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>2. Lab logbook</td><td>20%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>3. Examination</td><td>30%</td><td>✓</td><td>✓</td><td>✓</td><td></td><td></td><td></td></tr><tr><td>Total</td><td>100 %</td><td colspan="6"></td></tr></table> Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: Assignment and lab logbook: Individual continuous assessments, including lab exercises and logbooks, will enhance the understanding of the taught knowledge through practical experiments, reading and writing. Examinations: At the end of the subject, the student’s scientific report on practical experiments will be evaluated for the student’s understanding of wafer fabrication and experiment characterization.	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed						a	b	c	d	e	f	1. Assignment	50%	✓	✓	✓	✓	✓	✓	2. Lab logbook	20%	✓	✓	✓	✓	✓	✓	3. Examination	30%	✓	✓	✓				Total	100 %						
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Reading List and References	Textbooks • S.M Sze, VLSI Technology, McGraw-Hill,1988. • Gary S. May and Costas J. Spanos, Fundamentals of Semiconductor Manufacturing and Process Control, John Wiley & Sons, 2006. References • R. Rajsuman, System-on-a-Chip Design and Test (Artech House, London, 2000. • M. Burn and G. Roberts, An introduction to mixed-signal IC Test and Measurement, Oxford University Press, 2001.
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